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# Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising via Explicit Predicate Maps and Stable Fuzzy Allocation

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**ABSTRACT** Optical coherence tomography is read through the contrast between retinal layers and the sharpness of their boundaries. A denoiser is trained on a pixel error instead, so it can raise fidelity and still lose them. Nothing sits between the denoiser and the reader to say what a good scan looks like and repair where one falls short. We make that repair a problem of its own, solved after denoising with the denoiser frozen. Four things are new. Six clinical properties in closed form read the noisy input and the denoised output alone, each returning a score and a map of where it fails, so they run on any scanner. Five rules in fuzzy logic turn those maps into a per-pixel allocation whose response to a shift in evidence we bound by an exact, attained constant. A bounded proposal is admitted according to how many of three constraints hold, we prove when contrast cannot fall, and the gain the rules allocate traces to a named property and place. One architecture, one set of rules and one set of budgets serve four denoisers of different families, only the head that reads features resized, though each is fitted separately. All six measures rise on two of the four, five of six on the others, contrast by 3.5 to 11.3 percent, speckle suppression by 23.7 to 57.1 and signal to noise by 8.0 to 23.0, at 0.17 to 0.98 dB of fidelity. Carried unchanged to two other datasets, 40 of 48 improve.

**INDEX TERMS** Clinical quality assessment, fuzzy logic, image denoising, neuro-symbolic reasoning, optical coherence tomography.

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## I. Introduction

**O**PTICAL coherence tomography (OCT) is the standard non invasive way to image the retina [1]. Every OCT image carries speckle, a multiplicative disturbance that arises from coherent light scattering inside tissue [2]. Speckle blurs the border between retinal layers, lowers the contrast between tissue and background, and makes automatic layer segmentation less reliable [3].

Deep denoisers remove much of that disturbance, but they are almost always trained to minimize the squared error against a clean reference, so a denoiser can raise the peak signal to noise ratio (PSNR) of a scan while making it less useful to a clinician. Layer boundaries lose definition, the contrast between tissue and the dark vitreous falls, and thin structures fade into the background, none of which a PSNR number shows.

## A. Motivation

The gap we address is not a missing denoiser. It is that nothing sits between the denoiser and the clinical reader to state what a good OCT image should look like and to repair the ways a denoised one falls short. Another neural block would hide the reason for every change inside learned weights, while a clinician, a regulator or a reviewer who sees a modified scan needs to know which property was judged deficient, where, and how large a change was allowed. We therefore treat post-correction as its own problem, keeping the denoiser frozen and attaching a small wrapper written as named image properties, explicit rules over them and explicit limits on the size of any change.

## B. New ideas in this paper

The new ideas are four. Edge filters, gating maps and fuzzy operators are not among them.

- 1) Six clinical properties that need no clean image. Section III-B defines six properties computed from the noisy input and the denoiser output alone, so they run on any scanner at test time. Each returns a score for the image and a map of where it fails, both as formulas.
- 2) A rule layer whose decision cannot jump. Section III-C builds the allocation from five rules joined by a conjunction with bounded slope and proves that a shift in the evidence moves the decision by at most a constant the rule layer itself fixes, and Section V-I measures the response the images actually produce against it.
- 3) A safety layer that decides how much of a proposal to keep. Section IV-B states three constraints at three scales, and how many hold sets the fraction of the change applied. Section IV-C proves a margin condition under which contrast cannot fall, tested on every test image.
- 4) One design for four denoisers. Only the head that reads backbone features changes with the backbone, while the rules, budgets, thresholds and loss weights do not, and Section V-E carries the fitted wrapper to two other datasets with nothing refitted.

### C. Related work

Denoising of OCT images. Learned denoisers fall into three families, residual convolutional networks such as DnCNN [4], designs that gate or reweight channels such as NAFNet [5] and KBNNet [8], and transformers such as SwinIR [6]. Methods built for OCT address the scarcity of paired data directly, through triplet cross fusion [9] and sparse representation [11], although the earlier still needs one high quality scan of the same eye [10]. All are trained to reduce a pixel error, so the quantity they optimize is not the one a clinician reads, and work on quality assessment for OCT [12] measures clinical quality without acting on it. This paper closes that loop.

Confidence and uncertainty. Deep ensembles and their relatives [13] are the standard way to attach a confidence to a prediction, while in OCT the networks that segment layers [3] report an accuracy rather than an uncertainty, so a reader learns how often the model is right and never where it is unsure. We feed such a map into the correction decision, and Section V-H reports that the map we built is not a confidence in this sense.

Symbolic reasoning and expert networks. Neuro symbolic computing [14] couples a learned model with explicit rules, and differentiable fuzzy logic makes the coupling trainable because the connectives become smooth functions of continuous truth values [7]. In medical imaging it has generated spinal reports from a symbolic graph [20] and predicted patient deterioration with a probabilistic logic program over convolutional

TABLE 1. Symbols used in the paper, in order of first use.

| Symbol                                 | Meaning                                                   |
|----------------------------------------|-----------------------------------------------------------|
| $\mathbf{y}$                           | Noisy input B-scan                                        |
| $\mathbf{x}$                           | Clean reference B-scan                                    |
| $f_\theta$                             | Frozen denoising backbone with parameters $\theta$        |
| $\mathbf{b}$                           | Backbone output, $\mathbf{b} = f_\theta(\mathbf{y})$      |
| $\phi$                                 | Trainable parameters of the wrapper                       |
| $\mathbf{c}$                           | Cooperation map with values in $[0, 1]$                   |
| $\mathbf{u}$                           | Demand map, $\mathbf{u} = \mathbf{1} - \mathbf{c}$        |
| $s_i$                                  | Score of clinical property $i$ , a number in $[0, 1]$     |
| $\mathbf{m}_i$                         | Failure map of clinical property $i$ , values in $[0, 1]$ |
| $\tau_i$                               | Decision threshold for property $i$                       |
| $\Delta$                               | Raw multiplicative gain proposed by the gain network      |
| $\lambda$                              | Per pixel strength map with values in $[0, 1]$            |
| $\mathbf{a}$                           | Allocation map produced by the rule layer                 |
| $\alpha$                               | Gain after allocation, strength and clamping              |
| $\mathbf{g}$                           | Background gate that suppresses change in the background  |
| $\mathbf{v}$                           | Gated multiplicative change added to the backbone output  |
| $\delta_e$                             | Additive edge correction                                  |
| $\mathbf{q}$                           | Candidate image before the safety decision                |
| $\tilde{\mathbf{x}}$                   | Final output of the wrapper                               |
| $\mathcal{T}, \mathcal{B}$             | Sets of tissue pixels and background pixels               |
| $\mu_{\mathcal{T}}, \mu_{\mathcal{B}}$ | Mean value over tissue and over background                |
| $\sigma_{\mathcal{B}}$                 | Standard deviation over background                        |

features [21], and fuzzy membership functions have been embedded in networks that process OCT directly, guiding the segmentation of fluid lesions [22] and replacing learned feature extraction so that what the model attends to reads as a logical statement [23]. A parallel line builds an image model from cooperating experts, steered mixtures combining constant experts through soft gating kernels [24], also for OCT speckle removal [25], and restoration networks routing an input through experts of different capacity by its difficulty [26]. Outside imaging, evolving fuzzy models identify actuator dynamics [17], learned models join a physical forward operator in ultrasound tomography [18], and sensitivity to initial conditions is treated as a source of uncertainty [19]. All rest on the principle we rely on, that a learned component is easier to trust when part of the decision is written in a language a person can read. What is missing is a rule layer that acts on the output of a strong denoiser, names the properties it protects and limits how far it may move the image.

## II. Notation and Problem Statement

### A. Notation

Images are real valued arrays of height  $H$  and width  $W$  with values in  $[0, 1]$ . We write  $\mathbf{y}$  for the noisy B-scan that comes off the scanner and  $\mathbf{x}$  for the clean reference obtained by averaging many repeated scans of the same location. A pixel index is written  $p$ , so  $y_p$  is one pixel of  $\mathbf{y}$ .

The symbol  $\odot$  is the Hadamard product,  $(\mathbf{t} \odot \mathbf{t}')_p = t_p t'_p$ . Table 1 lists the symbols that recur across sections, and a symbol used in one place only is defined where it is introduced.

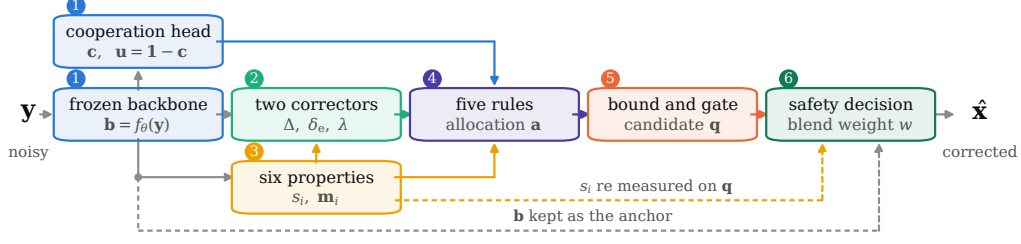


FIGURE 1. The six stages. A frozen backbone and a small head produce the image and the cooperation map. Two networks propose changes. Six properties are measured. Five rules allocate. The change is bounded and gated. Three constraints decide how much is kept.

### B. The optimization problem

The denoiser is given and is not changed. The task is to produce an image that beats  $\mathbf{b} = f_\theta(\mathbf{y})$  on named clinical measures, stays close to it everywhere, and can explain every change it makes. The only free quantities are the wrapper parameters  $\phi$ , namely the cooperation head, the gain network, the edge network, the strength predictor and the rule layer, while the backbone parameters  $\theta$  are held fixed throughout. Let  $\mathcal{D}$  be a set of paired scans and  $\hat{\mathbf{x}}(\phi)$  the output of the wrapper. We minimize the expected value of a weighted sum of six terms with fixed weights  $w_c, w_p, w_o, w_s$  and  $w_r$ ,

$$\min_{\phi} \mathbb{E}_{(\mathbf{y}, \mathbf{x}) \sim \mathcal{D}} \left[ \mathcal{L}_{\text{fid}} + w_c \mathcal{L}_{\text{clin}} + w_p \mathcal{L}_{\text{pred}} + w_o \mathcal{L}_{\text{coop}} + w_s \mathcal{L}_{\text{edge}} + w_r \mathcal{L}_{\text{reg}} \right], \quad (1)$$

where  $\mathcal{L}_{\text{fid}}$  keeps the output near the clean reference,  $\mathcal{L}_{\text{clin}}$  rewards improvement of the clinical measures over the backbone,  $\mathcal{L}_{\text{pred}}$  asks four property scores to rise above what the backbone already reaches,  $\mathcal{L}_{\text{coop}}$  ties the amount of correction to the estimated demand,  $\mathcal{L}_{\text{edge}}$  supervises the gradient field, and  $\mathcal{L}_{\text{reg}}$  collects the bounds on how large and how rough the change may be together with four terms that protect the background. Section V gives every term with its weight.

Three constraints are checked on every image and stated formally in Section IV-B. The aggregate clinical quality should not fall by more than a fixed tolerance, no single property should fall by more than a fixed tolerance while another rises, and the relative change from the backbone output should stay below a fixed bound. How many hold decides how much of the change is kept, so the rule is a majority test and not three hard guarantees. Every logical connective is a piecewise linear surrogate, so the objective is differentiable in  $\phi$  including the rule layer and is solved with AdamW [16]. The constraints are not relaxed into the objective. They are evaluated exactly, per image, at test time only, so the objective is optimized at  $\mathbf{q}$  and not at  $\hat{\mathbf{x}}$ , the acceptance rule not running while the wrapper is fitted. The two coincide whenever the rule accepts in full, which Section V-J reports is every test image.

### III. Clinical Properties and a Stable Rule Layer

Fig. 1 shows the method as six stages. Stage one asks where the backbone output can be left alone, two what change is available, three what is wrong and where, four where it should be applied, five how, and six whether it is allowed at all. This section carries the first four and proves that the decision cannot jump. Section IV carries the last two and proves when contrast cannot fall.

#### A. Stages one and two, the cooperation map and two bounded proposals

The backbone is frozen after training and gives  $\mathbf{b} = f_\theta(\mathbf{y})$ . The cooperation map  $\mathbf{c}$  says where  $\mathbf{b}$  can be left alone and where a change is wanted. A two layer network  $h_\phi$  reads a feature tensor  $\mathbf{z}$  from inside the backbone and produces it,

$$\mathbf{c} = \sigma \left( \frac{h_\phi(\mathbf{z}) + \beta_0}{\tau_0} \right), \quad (2)$$

where  $\sigma$  is the logistic function,  $\tau_0 > 0$  is a learnable temperature and  $\beta_0$  a learnable offset. The complementary demand map is  $\mathbf{u} = \mathbf{1} - \mathbf{c}$ .

Only this head depends on the backbone, since architectures expose feature tensors of different width. We read the first encoder stage of NAFNet and KBNet, the first convolution of DnCNN and the shallow feature convolution of SwinIR, and the head is a pair of convolutions sized to whichever tensor it is given, the widths listed in the supplementary file. Everything after this stage is shared, which is the precise sense in which the method is modular rather than architecture free.

Two bounded proposals. Two changes of deliberately different kinds are proposed.

The gain network proposes a multiplicative change. It runs at quarter resolution and emits a brightness channel through a rectifier, so it can only brighten, and a contrast channel through a hyperbolic tangent, so it can go either way. Their sum is scaled by  $\alpha_{\text{max}} = 0.15$  to give the raw gain  $\Delta$ . A multiplicative form suits speckle, which is itself multiplicative [2], so bright tissue receives a larger absolute change than dark background for the same relative adjustment.

The edge network proposes an additive change  $\delta_e$  from the intensity normalized residual, the edge magnitude of the backbone output, that output itself and an unsharp mask. It is bounded to  $\pm 0.08$  and scaled by a learned

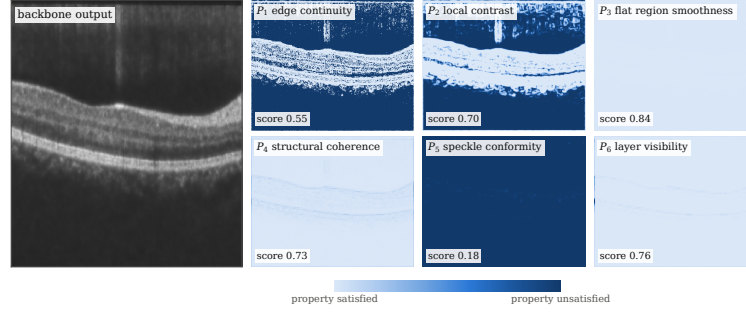


FIGURE 2. The six failure maps on one backbone output, with the score each property returns for the whole image. Dark means strong local evidence that the property is unsatisfied. The maps disagree, which is the point of keeping six. Edge continuity and local contrast fire along the layer boundaries and through the tissue between them, speckle conformity fires almost everywhere here and drives its low score, and the other three stay pale. The allocation of Equation (5) is built from these maps, so a reader can see which property asked for a change at any pixel.

factor. An additive form suits boundaries, which should be sharpened by the same amount in a bright layer as in a dark one.

A third small network predicts a per-pixel strength map  $\lambda$  with values in  $[0, 1]$  from the failure maps of Stage three. It modulates how much of the raw gain survives at each pixel.

#### B. Stage three, the six clinical properties

Each of the six properties names a way a denoised OCT image is known to fail and each is written as a formula so it can be recomputed by anyone. Table 2 gives, for every property, the reason it exists, its score  $s_i \in [0, 1]$  as a weighted sum of named subterms, its map  $\mathbf{m}_i \in [0, 1]^{H \times W}$  in closed form, where that map is large, and the threshold  $\tau_i$  that separates satisfied from unsatisfied, with every subterm defined beneath the table. A larger score means the property is better satisfied and a larger map value means stronger local evidence that it is not, and Fig. 2 shows all six maps on one image. None of these quantities needs a clean reference. They are computed from the noisy input and the backbone output only, so they remain available at test time on a scanner the method has never seen.

#### C. Stage four, the rule layer, and why its decision cannot jump

The rule layer decides where the gain proposal should act through five named rules, each casting a signed vote.

Each property score is first turned into a soft failure signal by comparing it with its threshold,

$$f_i = \sigma(\kappa(\tau_i - s_i)), \quad \kappa = 10, \quad (3)$$

so  $f_i$  is near one when property  $i$  is failing. The same soft comparison tests the cooperation map and the strength of a proposal against their own thresholds. Each corrector produces that strength as a per-pixel map from the backbone output and the failure maps it serves, large where its property is far below threshold, and it is multiplied by one half plus the demand map before the rules fire, so the same proposal looks stronger where the head reports low cooperation. Conjunction

uses the Lukasiewicz form  $\text{AND}(a, b) = \max(0, a + b - 1)$ , piecewise linear with a bounded slope, so each truth value is a Lipschitz function of the evidence beneath it. The constant of Theorem 1 follows from the allocation being affine in those truth values and does not depend on this particular conjunction.

The five rules follow. Rule one fires when the cooperation map is high and the proposal weak, and votes against changing the image. Rule two fires when the map is low and the proposal strong, and votes for. Rule three fires when a relevant property is failing, and votes for. Rule four fires when map and strength are both middling, and votes weakly for. Rule five fires when several correctors want the same pixel, and votes against. Written out, with  $\text{sg}(x, \tau) = \sigma(10(x - \tau))$ ,  $\text{sl}(x, \tau) = \sigma(10(\tau - x))$ ,  $c$  the cooperation map,  $\pi$  the proposal strength and  $s_{\min}$  the smallest property score the corrector reads,

$$\begin{aligned} r_1 &= \text{AND}(\text{sg}(c, \tau_{\text{hc}}), \text{sl}(\pi, \tau_{\text{hp}})), \\ r_2 &= \text{AND}(\text{sl}(c, \tau_{\text{lc}}), \text{sg}(\pi, \tau_{\text{hp}})), \\ r_3 &= \text{sl}(s_{\min}, \tau_{\text{pf}}), \quad r_4 = \text{AND}(m(c), m(\pi)), \\ r_5 &= \text{sg}(\sum_j \text{sg}(\pi_j, \tau_{\text{cf}}), 1.5), \end{aligned} \quad (4)$$

where  $m(x) = \text{AND}(\text{sl}(x, 0.7), \text{sg}(x, 0.3))$  selects the middle band and the sum in  $r_5$  runs over the correctors proposing at that pixel. The five thresholds  $\tau_{\text{hc}}$ ,  $\tau_{\text{lc}}$ ,  $\tau_{\text{hp}}$ ,  $\tau_{\text{p}}$  and  $\tau_{\text{pf}}$  are trainable, start at 0.40, 0.55, 0.30, 0.10 and 0.75, and move slightly during fitting. They are not the six property thresholds  $\tau_i$ , which take no gradient. In the reported configuration the allocation is formed for one multiplicative proposal, so the sum in  $r_5$  has a single term and  $r_5$  is zero everywhere.

The allocation at each pixel is the bounded weighted vote

$$\begin{aligned} a &= \underbrace{B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta)}_{\text{base, no rule firing}} \\ &\quad - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 + C_4 w_4 r_4 - C_5 w_5 r_5, \end{aligned} \quad (5)$$

where  $\beta$  is a learnable base logit,  $w_k = \sigma(\theta_k) \in (0, 1)$  are learnable rule weights and the budgets are  $B_{\min} = 0.25$ ,

$B_{\max} = 0.50, C_1 = 0.15, C_2 = 0.25, C_3 = 0.15, C_4 = 0.10$  and  $C_5 = 0.10$ .

The budgets satisfy  $B_{\max} + C_2 + C_3 + C_4 = 1$  and  $B_{\min} - C_1 - C_5 = 0$ . Since every truth value lies in  $[0, 1]$  and every weight in  $(0, 1)$ , these two identities are enough for Equation (5) to lie in  $[0, 1]$  for every input and every parameter value, so no clipping is ever needed. They are sufficient conditions and not a derivation of these particular values, whose relative sizes remain a design choice. Without them a clamp is needed, the allocation saturates at one end of it, as it did on every pixel of the submitted version of this work, the rule weights receive no gradient, and Theorem 1 is vacuous.

What stability means here. This is not stability in the sense of Lyapunov and we claim no fixed point. The quantity that must not behave erratically is the decision of the rule layer, a function of the five rule truth values, which are what shift when the backbone or the scanner changes.

**Definition 1** (Stability of the allocation). Let  $\mathbf{r} = (r_1, \dots, r_5) \in [0, 1]^5$  be the vector of rule truth values at one pixel and let  $a(\mathbf{r})$  be the allocation of Equation (5) at that pixel with the parameters held fixed. The allocation is said to be stable with constant  $L$  if

$$|a(\mathbf{r}) - a(\mathbf{r}')| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty \quad \text{for all } \mathbf{r}, \mathbf{r}' \in [0, 1]^5. \quad (6)$$

In words,  $L$  is the worst case ratio between a change in the decision and the change in the evidence that caused it. Without it the layer could swing from no correction to all of it because one score moved slightly.

**Theorem 1** (Allocation stability). The allocation of Equation (5) satisfies Definition 1 with

$$L = C_1 w_1 + C_2 w_2 + C_3 w_3 + C_4 w_4 + C_5 w_5, \quad (7)$$

where  $w_k = \sigma(\theta_k)$ . The constant is attained, so it is the exact Lipschitz constant and not merely an upper bound. Moreover  $L < C_1 + C_2 + C_3 + C_4 + C_5 = 0.75$  for every possible setting of the parameters. If a rule never fires its term drops out and the same argument gives the exact constant of the remaining sum, which Section V-I reports for rule five here.

**Proof:**

The allocation is affine in  $\mathbf{r}$  with coefficients  $\pm C_k w_k$ , so subtracting two values and applying the triangle inequality gives  $|a(\mathbf{r}) - a(\mathbf{r}')| \leq \sum_k C_k w_k |r_k - r'_k| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty$ . The bound is attained when each  $r_k - r'_k$  takes the sign of its coefficient, and  $L < 0.75$  because every  $C_k$  is positive and every  $w_k$  lies strictly inside  $(0, 1)$ . ■

**Corollary 1** (Behavior under a change of backbone). If replacing one backbone by another changes the rule truth values at a pixel by at most  $\epsilon$  in the maximum norm, the parameters unchanged, the allocation there changes by at most  $L\epsilon$ .

This concerns the routing rule alone, not whether image quality transfers, which Section V-E settles by measurement.

#### IV. Bounded Correction and Majority Based Acceptance

A. Stage five, applying the change

The admitted gain is the raw proposal modulated by the allocation and the strength map, then bounded,

$$\alpha = \text{clamp}(\Delta \odot \mathbf{a} \odot \boldsymbol{\lambda}, -0.20, 0.20). \quad (8)$$

A gate suppresses any change in the dark background,

$$\mathbf{g} = \sigma(20(\mathbf{b} - q_{0.15})), \quad (9)$$

where  $q_{0.15}$  is the fifteenth percentile of the backbone output of that image, a percentile rather than a fixed level so the gate works across scanners of different brightness. The second form, used where the selection of Section V-B chose it, replaces the percentile by the per image Otsu threshold and dilates the result with a maximum filter of side  $R$ ,

$$\mathbf{g} = \text{maxpool}_R(\sigma(20(\mathbf{b} - \text{otsu}(\mathbf{b})))), \quad R = 41, \quad (10)$$

which widens the admitted region to a band around tissue, because the edge branch earns its contrast and texture gains just outside the tissue boundary. Both gates read the detached backbone output and carry no gradient. Neither is ever exactly zero, since a logistic is strictly positive and a maximum filter preserves that. Over the test scans the smallest value Equation (10) reaches is 0.017 to 0.071 by image and Equation (9) 0.045 to 0.20, so the darkest background is attenuated by one to two orders of magnitude and not excluded, and the background measures of Section V-J are evidence rather than a property the gate guarantees.

The gated gain is applied multiplicatively to give the change  $\mathbf{v}$ , with the brightening part smoothed before it is added, which attenuates high frequency variation in that contribution,

$$\mathbf{v} = (\text{smooth}(\text{ReLU}(\alpha) \odot \mathbf{b}) + (\alpha - \text{ReLU}(\alpha)) \odot \mathbf{b}) \odot \mathbf{g}, \quad (11)$$

where smooth is a three by three mean filter, so a brightening is spread over a neighborhood while a darkening acts where it was proposed. The background is then smoothed. Writing  $\mathbf{u} = \mathbf{b} + \mathbf{v}$  for the image at that point,  $\beta = \frac{1}{2} \text{erode}_5(\mathbf{1} - \mathbf{g})$  for the blend weight and  $S(\mathbf{u})$  for a three by three mean taken over the masked pixels only, the smoothed image is  $\tilde{\mathbf{u}} = (\mathbf{1} - \beta) \odot \mathbf{u} + \beta \odot S(\mathbf{u})$ . The erosion keeps the blend away from the tissue border, and since  $\mathbf{g}$  is never one the weight is small rather than zero inside tissue. The edge proposal is added after this, and the result is clipped,

$$\mathbf{q} = \text{clamp}(\tilde{\mathbf{u}} + \delta_e \odot \mathbf{g}, 0, 1). \quad (12)$$

TABLE 2. The six clinical properties in closed form. Here  $e$  is the Sobel edge magnitude of  $\mathbf{b}$ ,  $\bar{e}$  its mean,  $b_e = \mathbf{1}[e > \bar{e}]$ ,  $\sigma_k$  and  $\text{Var}_k$  the local standard deviation and variance over a  $k \times k$  window,  $\bar{\sigma}$  the mean of  $\sigma_9$ ,  $v(\cdot)$  the depth profile averaged across width,  $\nabla_z$  the derivative along depth,  $\rho_h$  the one pixel horizontal autocorrelation,  $\text{sigm}$  the logistic function,  $\varepsilon$  a small constant, and  $cv$  the local coefficient of variation of the log residual. The named subterms are expanded in the supplementary file.

| $P_i$ | Property             | Score $s_i$                                                                               | Map $\mathbf{m}_i$                                                                                                                        | Where the map is large                                                 | $\tau_i$ |
|-------|----------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------|
| $P_1$ | edge continuity      | $0.4 c_{\text{cont}} + 0.3 c_{\text{efr}} + 0.3 c_{\text{ang}}$                           | $(1 - \bar{e})(1 - b_e)$ , $\bar{e} = e/(\max e + \varepsilon)$                                                                           | off detected edges, where a boundary should continue but does not      | 0.60     |
| $P_2$ | local contrast       | $0.20 s_{\text{range}} + 0.35 s_{\text{cnr}} + 0.20 s_{\text{var}} + 0.25 s_{\text{cov}}$ | $[\text{ReLU}(\bar{\sigma} - \sigma_9(\mathbf{b})) / (\bar{\sigma} + \varepsilon) + 0.3 \mathbf{1}[\sigma_9(\mathbf{b}) < 0.01]]_{[0,1]}$ | where $\sigma_9$ falls below the image mean, and in nearly flat tissue | 0.50     |
| $P_3$ | flat smoothness      | $0.6 q_{\text{var}} + 0.4 q_{\text{res}}$                                                 | $F \odot \sigma_5(\mathbf{b}) / (\max \sigma_5(\mathbf{b}) + \varepsilon)$                                                                | inside flat regions that still carry residual variation                | 0.75     |
| $P_4$ | structural coherence | $0.4 s_{\text{self}} + 0.3 s_{\text{reg}} + 0.3 s_{\text{vert}}$                          | $e(e) / (\max e(e) + \varepsilon)$                                                                                                        | where the edge map is itself rough, a sign of broken structure         | 0.50     |
| $P_5$ | speckle conformity   | $0.7 \text{mean}(\exp(- cv - cv^* /0.15)) + 0.3 \exp(-5 \rho_h(z) )$                      | $1 - \exp(- cv - cv^* /0.15)$                                                                                                             | where $cv$ departs from the value expected for OCT speckle             | 0.30     |
| $P_6$ | layer visibility     | $0.4 s_{\text{clr}} + 0.3 s_{\text{dep}} + 0.3 s_{\text{smo}}$                            | $\text{Var}_7(e) / (\max \text{Var}_7(e) + \varepsilon)$                                                                                  | where the local variance of the edge map is high across depth          | 0.70     |

B. Stage six, the safety decision

The candidate  $\mathbf{q}$  is not returned directly. Three constraints are checked on it.

Aggregate quality must not fall. Define for any image  $\mathbf{t}$  the energy

$$E(\mathbf{t}) = 1 - \frac{1}{6} \sum_{i=1}^6 s_i(\mathbf{t}) + 0.1 \sum_{i=1}^6 \max(0, 0.5 - s_i(\mathbf{t})), \quad (13)$$

which is small when the image is good and whose second term penalizes any property below one half, so a good average cannot hide one broken property. The constraint is  $E(\mathbf{q}) \leq E(\mathbf{b}) + 0.10$ .

No property may be sacrificed for another. The second constraint requires that at least one property improves by more than 0.01 and that no property falls by more than 0.10.

The image must not move far. The third constraint requires  $\|\mathbf{q} - \mathbf{b}\|_2 / \|\mathbf{b}\|_2 < 0.2$ .

Let  $n \in \{0, 1, 2, 3\}$  be the number of constraints that hold. The output is

$$\hat{\mathbf{x}} = (1 - w) \mathbf{b} + w \mathbf{q}, \quad w = \begin{cases} 1, & n \geq 2, \\ 0.05 + 0.10 \frac{n}{3}, & n < 2. \end{cases} \quad (14)$$

The candidate is accepted in full whenever any two constraints hold, so one violation does not by itself withhold the change. Below two the output falls back to the backbone while keeping five percent of the change when none holds and about eight when one does. This is a majority test on three measured conditions and not an enforcement of three bounds, and Section V-J reports how often each holds.

C. Sufficient conditions for non decreasing contrast

Write  $\text{CNR}(\mathbf{t}) = (\mu_{\mathcal{T}}(\mathbf{t}) - \mu_{\mathcal{B}}(\mathbf{t})) / \sigma_{\mathcal{B}}(\mathbf{t})$  for the separation between tissue and background divided by the variability of the background.

Theorem 2 (Contrast margin). Let  $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$  be the gain in tissue mean and let  $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$  bound the leakage into the background. Suppose the tissue and background sets are the same

for  $\mathbf{b}$ ,  $\mathbf{q}$  and  $\hat{\mathbf{x}}$ , that  $\sigma_{\mathcal{B}}(\hat{\mathbf{x}}) > 0$  and  $\sigma_{\mathcal{B}}(\mathbf{b}) > 0$ , so both ratios are defined, that the correction does not raise the background variability, so  $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$ , and that the bounded numerator  $(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})$  is non negative, which holds whenever the backbone already separates tissue from background and  $\eta_{\mathcal{T}} \geq \eta_{\mathcal{B}}$ . Then the output of Equation (14) satisfies

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})}{\sigma_{\mathcal{B}}(\mathbf{b})}. \quad (15)$$

In particular, whenever  $\eta_{\mathcal{T}} > \eta_{\mathcal{B}}$  the correction cannot reduce the contrast to noise ratio.

Proof:

The mean over a fixed pixel set is linear, so  $\mu_{\mathcal{T}}(\hat{\mathbf{x}}) = \mu_{\mathcal{T}}(\mathbf{b}) + w\eta_{\mathcal{T}}$  and  $\mu_{\mathcal{B}}(\hat{\mathbf{x}}) \leq \mu_{\mathcal{B}}(\mathbf{b}) + w\eta_{\mathcal{B}}$ , which bounds the numerator of  $\text{CNR}(\hat{\mathbf{x}})$  below by the numerator of Equation (15). On the background  $\hat{\mathbf{x}}$  is the same convex combination pixel by pixel, so Minkowski's inequality on the centered values gives  $\sigma_{\mathcal{B}}(\hat{\mathbf{x}}) \leq (1 - w)\sigma_{\mathcal{B}}(\mathbf{b}) + w\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$ . Dividing the two bounds gives the claim, and  $w > 0$  in every branch of Equation (14). ■

Three parts of the design encourage these hypotheses rather than enforce them. The gate of Equation (9) keeps  $\eta_{\mathcal{B}}$  small by suppressing change in the dark region, the brightness branch cannot be negative, and the background smoothing lowers  $\sigma_{\mathcal{B}}(\mathbf{q})$  rather than raising it. None settles the matter, since the net tissue change also passes through the contrast branch, the edge branch and a final clipping, any of which can turn it negative. Section V-I therefore measures each hypothesis rather than assuming it.

Both guarantees are deliberately narrow. Theorem 1 bounds how far the routing decision can move, not whether it is correct. Theorem 2 concerns contrast alone. Neither promises that the output is closer to the clean reference, which Section V-D shows it usually is not. They concern the wrapper, not the denoiser underneath it, and Section V-I tests both on real images.

## V. Training, Tests and Results

### A. The six loss terms

Let  $\mathbf{x}$  be the clean reference,  $\mathbf{b}$  the backbone output and  $\hat{\mathbf{x}}$  the wrapper output. The objective of Equation (1) has the following terms.

**Fidelity.** Let  $D = \text{PSNR}(\mathbf{b}) - \text{PSNR}(\hat{\mathbf{x}})$  be the drop against the backbone in decibels and  $D_s$  the corresponding drop in the structural similarity index [15], both taken against the clean reference  $\mathbf{x}$ . The symbol is  $D$ , since Equation (8) already uses  $\Delta$  for the gain map. Neither is charged until it exceeds a dead zone,

$$\mathcal{L}_{\text{fid}} = 5 [D - d]_+ + 15 [D - d - 0.4]_+^2 + 20 [D_s - 0.005]_+, \quad (16)$$

where  $[\cdot]_+$  is the positive part. The term is flat while the drop stays below  $d$  decibels, linear for the next four tenths and quadratic beyond, so a small loss of fidelity is free and a large one expensive. The dead zone  $d$  is one of the two quantities selected per backbone in Section V-B.

**Clinical improvement.** This term penalizes falling short of a target on one side, with a symmetric term for three of the nine that also penalizes overshooting, so a gain in one measure is discouraged from being paid for by a collapse in another. Let  $q_m$  be quantity  $m$  on the wrapper output and  $t_m = \kappa_m q_m(\mathbf{b})$  its target, the backbone value scaled by a fixed factor. Writing  $\varepsilon_m = t_m - q_m$ ,

$$\mathcal{L}_{\text{clin}} = \frac{1}{\sum_m w_m} \sum_{m \in M} w_m \left( 3[\varepsilon_m]_+ + [\varepsilon_m]_+^2 + \frac{1}{2} s_m \varepsilon_m^2 \right). \quad (17)$$

The set  $M$  holds nine quantities, the three preservation scores of local contrast, boundary gradient and texture, then EPI, CNR, BS, ENL, SNR and the ratio of the backbone squared error to the wrapper squared error, each defined in the supplementary file with its window, mask and reduction. Their weights  $w_m$  are 1, 5, 1, 15, 40, 5, 8, 40 and 10 and their factors  $\kappa_m$  are 1.12, 1.12, 1.00, 1.05, 1.20, 1.12, 1.10, 1.15 and 0.87, the last allowing about six tenths of a decibel. The indicator  $s_m$  is one for ENL, SNR and the squared error ratio, a symmetric pull holding those three near target. None of the nine is used at test time.

**Preservation scores.** This term acts on four scores  $g_j$  separate from the six properties of Table 2, comparing corrected against backbone on local contrast, the boundary gradient, the local variance carrying texture and the edge magnitude. With  $\delta_j = g_j(\hat{\mathbf{x}}) - g_j(\mathbf{b})$ , the backbone value fixed,

$$\mathcal{L}_{\text{pred}} = \sum_{j=1}^4 v_j \left( 3[-\delta_j]_+ - 2.5[\delta_j]_+ + 2 \left[ \frac{1}{2} - g_j(\hat{\mathbf{x}}) \right]_+ \right), \quad (18)$$

where  $v_j$  is 3, 2, 2 and 1 in that order. A drop costs more than a rise earns, and any score below one half is penalized again. The thresholds  $\tau_i$  play no part here. They enter Equation (3), evaluated in a module whose

output reaches the objective only through detached quantities, so no gradient path reaches  $\tau_i$  and all six hold their Table 2 values throughout training. This is not true of the rule layer as a whole. The base level  $\beta$  and the rule weights  $\theta_k$  of Equation (5) are differentiable and do receive gradients, which Section V-I measures parameter by parameter.

**Cooperation.** The wrapper should act where the backbone leaves room and stay quiet elsewhere. We take the Pearson correlation  $\rho$  between the demand map  $\mathbf{u}$  and  $\boldsymbol{\pi}$ , the summed proposal strength the rule layer reads, and penalize disagreement,

$$\mathcal{L}_{\text{coop}} = \frac{1}{2} \text{ReLU}(1 - \rho(\mathbf{u}, \boldsymbol{\pi})). \quad (19)$$

Each map is first rescaled to  $[0, 1]$  by its own minimum and maximum, so the term is invariant to a positive affine change of either but not to a negative one, which reverses the sign. The correlation is undefined for a constant map, where the implementation returns a zero loss, the one place this term is not differentiable.

**Edge supervision.** The additive branch is trained to reproduce the gradient that the backbone lost, measured with the Sobel operator  $\nabla$  and weighted toward the places where the clean reference has an edge. Let  $\mathbf{w}$  be the clean Sobel magnitude divided by its mean and clipped at ten,  $\tilde{\mathbf{w}}$  its square normalized again to unit mean, and  $\tilde{\mathbf{g}}$  a Sobel magnitude divided by its own maximum. With  $\delta = [1.05 \text{EPI}(\mathbf{b}) - \text{EPI}(\hat{\mathbf{x}})]_+$ ,

$$\begin{aligned} \mathcal{L}_{\text{edge}} = & 10 \left\| \sqrt{\tilde{\mathbf{w}}} \odot (\nabla \hat{\mathbf{x}} - \nabla \mathbf{x}) \right\|_2^2 \\ & + 10 \left\| \sqrt{\tilde{\mathbf{w}}} \odot (\tilde{\mathbf{g}}(\hat{\mathbf{x}}) - \tilde{\mathbf{g}}(\mathbf{x})) \right\|_2^2 + 45 (3\delta + \delta^2). \end{aligned} \quad (20)$$

The first part is the gradient the correction must restore, the second compares normalized edge magnitude, and the third penalizes an edge preservation index below five percent above the backbone.

**Bounds and background protection.** The last term keeps the change small, smooth and confined. Let  $A$  be the mean absolute change over the upper half of the image, named  $A$  because  $a$  is the allocation and  $\mathbf{g}$  the gate, let  $e$  be the mean absolute output of the edge branch, and  $\mathbf{m}$  a soft mask below the fifteenth percentile of the backbone output. Here  $\sigma_R$  and  $\mu_U$  are taken over the lower quarter and the upper half of the frame, fixed bands and not the anatomical sets  $\mathcal{T}$  and  $\mathcal{B}$  of Theorem 2,

$$\begin{aligned} \mathcal{L}_{\text{reg}} = & \min(40[0.002 - A]_+, 5) \\ & + \min(200[A - 0.006]_+^2, 5) \\ & + 2 \|\nabla \boldsymbol{\alpha}\| + 30[e - 0.06]_+^2 \\ & + 80[\sigma_R(\hat{\mathbf{x}}) - 0.75 \sigma_R(\mathbf{b})]_+ \\ & + 100 \left[ \frac{\sigma_R(\hat{\mathbf{x}})}{\sigma_R(\mathbf{b})} - 0.95 \right]_+ \\ & + 150 \left[ 0.99 - \frac{\mu_U(\hat{\mathbf{x}})}{\mu_U(\mathbf{b})} \right]_+ + 2 \|\hat{\mathbf{x}} - \mathbf{b}\| \odot \mathbf{m}, \end{aligned} \quad (21)$$

where a bar denotes the mean over pixels. Every squared norm in Equations (20) and (21) is likewise a mean

over pixels and over the batch, not a sum, so no term changes scale with the crop size. The floor discourages the wrapper from learning to do nothing and the ceiling from overshooting fidelity, the third keeps the admitted gain  $\alpha$  of Equation (8) smooth, not the allocation, the fourth bounds the edge branch, and the last three penalize a rise in background noise or a fall in tissue brightness.

#### B. Schedule, selection and constants

Data and optimization schedule. Training uses paired data, a noisy scan and a clean reference averaged over repeated acquisitions. The reference enters the fidelity and clinical terms and supervises the edge branch. Testing uses no clean image, since the six properties read the noisy input and the backbone output alone. The backbone is trained first with the squared error and frozen. The wrapper follows in two stages, every term but fidelity active at first so the correctors learn a useful direction, then fidelity from the midpoint so they learn how far they may go. Both stages use AdamW with weight decay  $10^{-4}$  on  $96 \times 96$  crops, in batches of sixteen and of eight for DnCNN and SwinIR, for ten epochs in all, with base learning rates of  $2 \times 10^{-4}$  for the correctors,  $5 \times 10^{-4}$  for the cooperation head and the strength predictor and  $3 \times 10^{-4}$  for the rule layer. The supplementary file restates the procedure as pseudocode.

The normalization statistics of  $f$  are held fixed. DnCNN is the one backbone carrying batch normalization, and left in training mode its buffers move, so the reference its clinical change is measured against would drift while the correction was fitted to it.

How the operating point was chosen. The terms of Equation (1) carry fixed relative weights, the same for every backbone and seed. The clinical term dominates at 6, the property term is 1, the cooperation term is 0.2, the edge and regularizer terms are 1 since their internal coefficients carry their scale, and the fidelity term is 0 in the first stage and 1 in the second, which is what the two stage schedule means. The parser also accepts ten weight flags, none of which reaches the loss, two being printed at start up, which is the only thing that made any look active. The release notes name all ten. Two quantities are selected per backbone on validation, the dead zone  $d$  of Equation (16) and the gate of Equation (9). The grid crosses the two gate rules with dead zones of 0.3, 0.6 and 0.9 dB and the tissue-bounded gate with radii of 41 and 61 pixels, everything else held at one value, each setting trained at three seeds and scored on validation. It chose the tissue-bounded gate at radius 41 with a dead zone of 0.6 dB for NAFNet and the percentile gate at 0.9 dB for the other three. A setting is admissible only if it clears its bounds at its worst seed, and among those we take the largest selection score on the weakest clinical measure, the mean over seeds less one standard error, relaxing

the bounds one step at a time when none clears them. NAFNet and SwinIR were admitted at the strictest level. No setting cleared any level for DnCNN or KBNet, so their reported setting is the least bad of the grid rather than one that met a stated bound. Those two are also the weakest on test, DnCNN being the only backbone whose boundary sharpness falls and KBNet the only one whose edge preservation is flat, at +0.0005 percent with one of its three seeds negative. The supplementary file gives the bound and the four levels. Every source of randomness is seeded, up to the nondeterminism of the convolution kernels, and the checkpoints chosen on validation are the ones scored on test.

Initialization and every constant. Every constant appears with its value where it acts and has one of four origins. Fixed structural choices, never tuned, are the sharpness  $\kappa = 10$  of Equation (3), the steepness 20 of both gates, the window sizes, the quarter resolution of the gain network, the bound  $\pm 0.08$  of the edge branch, the thresholds  $\tau_i$  of Table 2, and the budgets  $B$  and  $C$ . Set once on validation and never revisited are  $\alpha_{\max} = 0.15$ , the gain bound  $\pm 0.20$ , the gate percentile 0.15 and radius 41, the three tolerances of Section IV-B, the half strength background blend and the learning rates. Selected in the grid of Section V-B, on the validation split only, are the dead zone  $d$  and the gate rule. Learned by the objective are  $\beta$ ,  $\theta_1$  to  $\theta_4$ ,  $\tau_0$ ,  $\beta_0$  and the scale of the edge branch. The fifth rule weight  $\theta_5$  is declared learnable and is not, for the reason Section V-I gives.

The wrapper is not fitted from scratch. Each run restores the whole wrapper, the rule layer included, from the corrector of the submitted version for that backbone and trains it for ten epochs with a fresh optimizer. Default initialization describes those earlier models, not the runs reported here. Nothing was chosen on the test split.

#### C. Setup

Data. PKU37 [9] provides 37 cross sections of  $640 \times 640$  pixels from a spectral domain scanner, references averaged over 50 frames, split by clean image identity into 27, 5 and 5 to give 1388, 173 and 173 pairs. It is the only training domain. Duke17 [10] provides 17 paired B-scans of  $970 \times 520$  from another institution, of which we use the 16 the released loader reads, and Duke2013 [11] the 18 volumes of  $900 \times 450$  of its test split. Both Duke sets come from one manufacturer and the 2013 evaluation reuses some 2012 material, so they are two external datasets and not two independent scanners.

Transfer protocol. Every model is trained once on PKU37 and applied to the Duke sets directly, with nothing fitted, recalibrated or selected on Duke data beyond the per image quantity the gate computes by the same rule as on PKU37. The original submission

TABLE 3. PKU37 test set, 173 images. Every cell is the change the wrapper makes to its frozen backbone, fidelity in decibels and the six clinical measures as a percentage, the mean over three seeds, bold where that mean exceeds the seed spread. The backbones alone read 30.75, 30.68, 29.87 and 32.04 dB in row order. Table S6 of the supplementary file gives the spread of each of the six clinical cells.

| Backbone | $\Delta$ PSNR    | $\Delta$ CNR  | $\Delta$ TCI  | $\Delta$ EPI | $\Delta$ BS  | $\Delta$ ENL  | $\Delta$ SNR  |
|----------|------------------|---------------|---------------|--------------|--------------|---------------|---------------|
| NAFNet   | -0.18 $\pm$ 0.02 | <b>+3.49</b>  | <b>+1.91</b>  | <b>+1.20</b> | <b>+3.89</b> | <b>+23.74</b> | <b>+11.10</b> |
| DnCNN    | -0.98 $\pm$ 0.21 | <b>+11.30</b> | +0.99         | <b>+4.14</b> | -1.44        | <b>+57.14</b> | <b>+22.95</b> |
| SwinIR   | -0.17 $\pm$ 0.06 | <b>+7.33</b>  | <b>+11.45</b> | <b>+2.68</b> | <b>+1.82</b> | <b>+28.32</b> | <b>+11.54</b> |
| KBNet    | -0.90 $\pm$ 0.08 | <b>+6.93</b>  | <b>+5.92</b>  | +0.00        | <b>+2.58</b> | <b>+25.30</b> | <b>+7.95</b>  |

instead adapted the wrapper for fifteen further epochs on the other subjects of the target set, leave one subject out, keeping in each fold the epoch scoring best on the subject it reports. Those numbers are selected on the data they describe, so the supplementary file keeps them as a historical diagnostic only.

Measures. Peak signal to noise ratio against the clean reference measures fidelity. Six clinically motivated measures, given as percentage change from backbone to corrected output, are the contrast to noise ratio of tissue against background, the tissue contrast index for dynamic range in tissue, the edge preservation index for agreement with the reference edge map, boundary sharpness for the gradient across layer boundaries, the equivalent number of looks for speckle suppression, and the signal to noise ratio against the noise floor. Each is computed per image then averaged, never pooled. The 173 pairs come from five clean image identities, so every count below describes repeated acquisitions of five anatomies and not a population.

#### D. In distribution results

Table 3 reports the PKU37 test set, Fig. 4 the corrected scans at full resolution and Fig. 3 the band where the correction is largest. The wrapper raises contrast on every backbone, by 3.5 to 11.3 percent, and the speckle measure by 23.7 to 57.1 percent, for 0.17 dB of fidelity at best and under one decibel at worst. Twenty one of the 24 backbone by measure pairs improve by more than one seed spread, one improves inside it, one is flat and one falls, the boundary sharpness of DnCNN. The gain holds image by image. Contrast rose on all but twelve of the 2076 image and seed combinations, none by as much as half a percent, and it is largest where the backbone leaves most to repair.

#### E. Transfer to other datasets

Table 4 carries the same four models to Duke17 and Duke2013 with nothing refitted. All six measures stay positive on both sets for NAFNet and SwinIR at most 0.35 dB in the reported means, and 40 of the 48 backbone by measure by set pairs are positive, the eight exceptions being the tissue contrast index and boundary sharpness of DnCNN and the edge preservation and signal to noise

TABLE 4. Duke17 and Duke2013, the models of Table 3 applied with no parameter updated, no threshold recalibrated and no statistic of the target images used to select anything. Columns and seeds as in Table 3. The backbones alone read 27.03, 26.89, 27.06 and 27.59 dB on Duke17 and 27.33, 27.21, 26.89 and 28.12 dB on Duke2013.

| Backbone | $\Delta$ PSNR    | $\Delta$ CNR  | $\Delta$ TCI  | $\Delta$ EPI | $\Delta$ BS  | $\Delta$ ENL  | $\Delta$ SNR  |
|----------|------------------|---------------|---------------|--------------|--------------|---------------|---------------|
| Duke17   |                  |               |               |              |              |               |               |
| NAFNet   | -0.17 $\pm$ 0.01 | <b>+7.51</b>  | <b>+4.00</b>  | <b>+1.37</b> | <b>+4.15</b> | <b>+21.35</b> | <b>+5.43</b>  |
| DnCNN    | -0.29 $\pm$ 0.05 | <b>+12.19</b> | -1.47         | <b>+3.49</b> | -2.24        | <b>+39.42</b> | <b>+13.55</b> |
| SwinIR   | -0.19 $\pm$ 0.05 | <b>+8.68</b>  | <b>+13.26</b> | <b>+2.69</b> | <b>+1.90</b> | <b>+29.46</b> | <b>+11.60</b> |
| KBNet    | -0.51 $\pm$ 0.05 | <b>+5.34</b>  | <b>+7.19</b>  | -0.50        | <b>+1.61</b> | <b>+7.49</b>  | -1.81         |
| Duke2013 |                  |               |               |              |              |               |               |
| NAFNet   | -0.35 $\pm$ 0.01 | <b>+7.37</b>  | <b>+1.41</b>  | <b>+0.85</b> | <b>+0.51</b> | <b>+28.80</b> | <b>+9.29</b>  |
| DnCNN    | -0.42 $\pm$ 0.05 | <b>+12.73</b> | -2.21         | <b>+2.99</b> | -5.85        | <b>+49.91</b> | <b>+17.59</b> |
| SwinIR   | -0.19 $\pm$ 0.06 | <b>+9.26</b>  | <b>+11.86</b> | <b>+2.55</b> | <b>+2.30</b> | <b>+30.90</b> | <b>+13.31</b> |
| KBNet    | -0.59 $\pm$ 0.04 | <b>+7.16</b>  | <b>+4.83</b>  | -0.35        | <b>+3.06</b> | <b>+8.70</b>  | -0.90         |

ratio of KBNet. Contrast rises on every backbone and both sets.

#### F. Which properties and which constants matter

One property at a time is switched off at test time, its score set to the satisfied value and its failure map to zero, and the test set re evaluated. Only  $P_5$ ,  $P_1$  and  $P_2$  move any measure by more than a quarter point, costing 3.19, 1.72 and 0.74 points of the speckle gain, while the other three change nothing by more than 0.23, each removed on its own. We keep all six, since they are the vocabulary of the rule layer and of the explanation, and one that rarely fires here can fire on another scanner.

We then vary the base level and the two rule weights that carry the allocation, and replace the Lukasiewicz conjunction by the product and the minimum, budgets held fixed. None of the fourteen variations turns a measure negative, contrast staying between 3.5 and 4.5 percent and fidelity between 0.16 and 0.23 dB. Only the base level has a visible effect, lowering it raising contrast and costing fidelity together, while the two weights move the result by under a fifth of a point.

G. What each component contributes, and how it compares Table 5 removes one component at a time from the selected NAFNet model. Replacing the rule layer by a uniform allocation cuts the contrast gain from 3.7 to 0.8 percent and the tissue contrast index from 2.4 to 0.4 while doubling the fidelity cost. The gain of this trained model therefore depends on its allocation, which is not the same as isolating the rule design from the capacity of the correctors, and a matched comparison would need a second model trained from scratch. Removing the edge branch turns both negative, removing the cooperation map costs about a third of the contrast gain, and removing background smoothing leaves contrast and the two edge measures almost intact while the speckle measure collapses from 23.3 to 5.8 percent.

The lower block spends a similar budget three other ways. The first is the same architecture, objective and ten epochs with the rule layer replaced by a uniform allocation throughout training. It was fitted before the

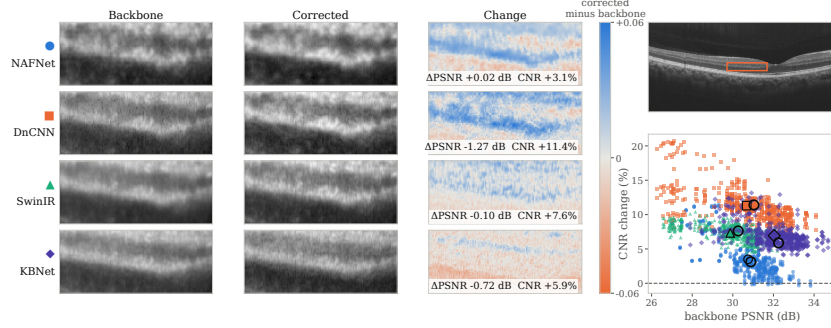


FIGURE 3. The band where the NAFNet correction is largest, 56 by 136 pixels, marked on the thumbnail, one row per backbone. The eight gray cells share one display range set to the values they contain, so nothing is clipped, and the signed change uses a fixed scale of plus or minus 0.06. The scatter places this image, ringed, among all 173 test images at three seeds for four backbones, with the twelve NAFNet points below zero marked.

TABLE 5. The seed 0 checkpoint of the selected NAFNet model on the same 173 test images, not a mean over seeds. Above, one component of the wrapper removed at a time. Below, three other ways of spending a similar budget on the same backbone output. Columns as in Table 3.

| Variant                                                        | $\Delta$ PSNR (dB) | $\Delta$ CNR | $\Delta$ TCI | $\Delta$ EPI | $\Delta$ BS | $\Delta$ ENL | $\Delta$ SNR |
|----------------------------------------------------------------|--------------------|--------------|--------------|--------------|-------------|--------------|--------------|
| One component removed                                          |                    |              |              |              |             |              |              |
| Full method                                                    | -0.17              | +3.71        | +2.42        | +1.21        | +4.16       | +23.34       | +10.74       |
| No rule layer                                                  | -0.38              | +0.83        | +0.38        | +0.99        | +4.27       | +25.45       | +10.44       |
| No edge branch                                                 | -0.37              | -2.71        | -6.54        | +0.55        | +0.65       | +20.28       | +7.30        |
| No cooperation map                                             | -0.15              | +2.51        | +1.15        | +1.10        | +3.87       | +23.84       | +10.61       |
| No bg smoothing                                                | -0.18              | +3.03        | +6.88        | +0.92        | +3.99       | +5.75        | +2.85        |
| Alternatives of similar complexity on the same backbone output |                    |              |              |              |             |              |              |
| Uniform allocation                                             | -0.42              | +4.11        | +11.58       | +0.71        | +3.83       | +16.18       | +5.80        |
| Unsharp masking                                                | -0.67              | +0.63        | +39.45       | -3.59        | +25.13      | -35.31       | -20.57       |
| Adaptive equalization                                          | -4.43              | +6.48        | +53.03       | -3.45        | +24.62      | -38.60       | -25.13       |

gate search, so it carries the percentile gate and not the tissue-bounded gate NAFNet was selected with, matched in architecture and not configuration. It reaches a larger contrast gain and a much larger tissue contrast index for 0.42 dB against 0.17, and is worse on four of the six measures, so the two are not ordered and this stays exploratory. The other two are classical operators tuned on the validation split. Both reach a tissue contrast index far above anything the wrapper does and both fail elsewhere, lowering edge preservation by about 3.5 percent and the speckle measure by about 36 percent while raising boundary sharpness by about 25 percent by sharpening noise, and adaptive equalization costs 4.4 dB.

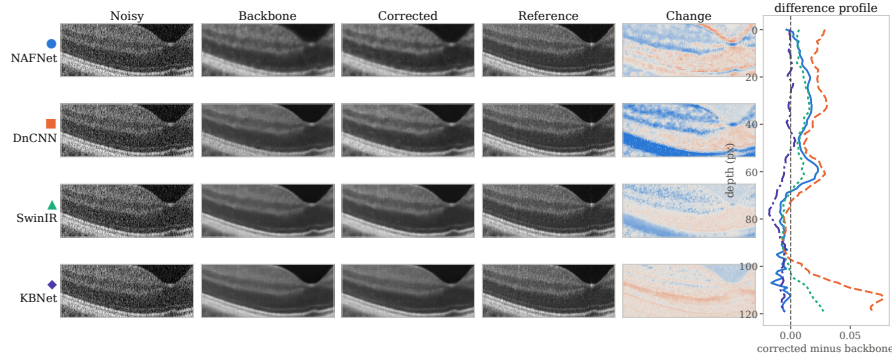
#### H. Does the cooperation map predict error

Called a confidence map in the original submission, this component was tested against the absolute error of the backbone output over 173 images and does not predict it. The quantity tested is the demand map  $\mathbf{u} = \mathbf{1} - \mathbf{c}$ , so one tracking backbone error would give a positive rank correlation. The measured value is -0.258, ordered the wrong way rather than uninformative, and removing the pixels it ranks least reliable raises the remaining error to 1.88 times its start where an oracle ranking reaches 0.07. It is useful even so, since replacing it by a constant lowers the contrast gain from +3.71 to +2.51 percent and worsens five of six measures. No target in the objective calibrates the map to backbone error, so we call it a cooperation map.

#### I. Do the theory and the rule layer hold up on real data

Which parameters actually train. A varying allocation map does not by itself show the rule weights were learned, since fixed coefficients on varying inputs look the same. We ran one training step on the selected model and read the gradient of every rule layer parameter. The base level  $\beta$  and the weights of rules one to four carry a gradient, and their values in the four checkpoints differ from what they were restored with, by as much as 1.49 in  $\beta$ . The five rule thresholds of Equation (4) also move. Rule five does not, since  $r_5 = 0$  when one corrector proposes, so  $-C_5 w_5 r_5$  vanishes for any  $w_5$ . The property thresholds  $\tau_i$  have no gradient path either, which is why they hold their literature values. Two constants follow, each exact on its own domain. On  $[0, 1]^5$  Theorem 1 gives 0.6014, and on  $[0, 1]^4 \times \{0\}$ , the fifth coordinate held where the method leaves it, 0.5489, which bounds any smaller set of attainable vectors.

Whether the conditions hold. We perturb the rule truth values and compare the ratio of Equation (6) with the constant of Equation (7), and measure the tissue gain, the background leakage and the ratio of background standard deviations against the hypotheses of Theorem 2. The bound holds with room to spare, 0.5489 against a largest measured ratio of 0.4679. The constant being exact on its domain, that gap is not slack in the proof, only perturbations that did not attain the worst case direction. The allocation map spans 0.556 to 0.793 with mean 0.726, and 0.00 percent of pixels sit at either end of its clamp, direct evidence that the rule layer does work after the repair of Section III-C.



**FIGURE 4.** The same PKU37 test image at full resolution, a 120 by 300 pixel region spanning the retinal layers, one row per backbone. Columns are the noisy input, the frozen backbone output, the corrected output, the clean reference, and the signed change, blue where the wrapper raised the intensity and orange where it lowered it. The change concentrates on the layer boundaries and the tissue between them. The right panel averages it across the width, each curve in the color of its row.

The margin condition holds on 139 of 173 images and the ratio condition on 94, and both together on 60. Those 60 are the images on which the two conditions we measure hold jointly, not a count on which the whole theorem applies, since its other hypotheses are properties of the decomposition and not quantities we evaluate per image. Where a hypothesis fails the theorem predicts nothing, and the contrast change there does not differ systematically in sign, so the conditions are sufficient and not necessary. Both leakage terms sit far below the mean absolute change over the image, 0.0067, at -0.0007 in tissue and -0.0017 in background, with the per image sign the direction the decomposition assumes on 137 and 107 of 173.

#### J. Safety, and whether the correction invents structure

A correction that raises contrast could sharpen a boundary that is not there or erase a faint one that is, and the clean reference measures both. Two move against it. An edge appears where the reference has none on 0.48 percent of pixels after correction against 0.45 for the backbone, a rise on 127 of 173 images, and the retention of structures in the lowest decile of reference contrast falls from 40.5 to 37.6 percent, on 168. We give these as effect sizes and not as tests, since the 173 acquisitions come from five anatomies and a paired test over them would not describe a population. The second is the one that matters, a fall of 2.9 points of retention, about 7.2 percent of where it starts. Neither is zero by construction, since the clamp bounds how far a pixel may move and not how many cross an edge threshold, and Section VI lists both. What separates the wrapper from a sharpening filter is Table 5, where the classical sharpeners lose edge preservation and the speckle measure while it gains on both.

Three checks come out in its favor. The two layer boundaries barely move, the inner limiting membrane shift falling from 4.73 to 4.38 pixels and the retinal pigment epithelium shift changing from 9.93 to 10.03. Neither is an equivalence test, and we ran none, so

this is an absence of measured movement and not a demonstration that anatomy is preserved. Dark fluid like regions occupy 1.6 percent of the image with a mean absolute change of 0.0022 inside them against 0.0067 overall, and the largest per image mean over the test set being 0.0178, a mean over pixels first and a maximum over images, not a single pixel extreme. The correction is quieter where a dark region carries the finding. Boundary sharpness is not worsened on 104 of 173 images. Of the three constraints of Section IV-B, energy holds on 173 and bounded change on 173, and only the no sacrifice constraint ever fails, on 173 minus 135. None falls below two, so the candidate is returned in full everywhere and the layer never acted on this test set.

#### K. Cost of moving to a new backbone

A new backbone means refitting the whole wrapper, not only its head. The trainable count is 168 630 shared parameters plus a head following the width of the feature tensor it reads, 4659 for KBNet, 7263 for NAFNet, 85 908 for SwinIR and 234 273 for DnCNN. Against frozen backbones of 7 583 521, 7 009 881, 6 939 289 and 7 029 013 parameters that is 2.3, 2.5, 3.7 and 5.7 percent, so the shared part is constant and the cost of attaching it follows how wide a tensor the backbone exposes. These times are for refitting from the warm start above, not for building a wrapper from nothing. Every backbone is fitted on the same 1388 pairs for the same ten epochs, so only the wall clock and the seed spread vary. One seed takes 2.0 minutes for NAFNet, 2.5 for KBNet, 3.9 for DnCNN and 8.3 for SwinIR on one RTX 5070 Ti, and the fidelity spread runs 0.02, 0.08, 0.21 and 0.06 dB in that order.

## VI. Discussion

### A. Why fidelity falls, and when that is the right trade

Fidelity falls in the mean on all four backbones, from 0.17 dB on SwinIR to 0.98 dB on DnCNN, and the reason is structural. A denoiser trained on squared error already sits near a local optimum of that error, so a change that

raises contrast or sharpens a boundary generally moves away from it. It is not universal image by image. The NAFNet panel of Fig. 3 gains 0.02 dB while its contrast rises, so what we report is an observed average and not a law, an average cost under one decibel for the quantities a grader reads.

Where the gain lands follows what each backbone leaves undone rather than how good it is overall. On boundary sharpness NAFNet starts furthest below the clean reference and gains most, while DnCNN starts above it and is the only one that falls. On edge preservation KNet starts highest and is the only one left flat. SwinIR is the instructive case, since windowed attention smooths hard inside a window, leaving the most room on contrast and paying the least fidelity. A deployment that cannot afford the payment lowers  $\alpha_{\max}$ , which attenuates the multiplicative branch but leaves the edge branch and the background operation active, so only bypassing the wrapper returns the backbone output exactly.

## B. Limitations

The correction is bounded, so a backbone that destroyed a structure cannot be made to recover it. Two safety measures move against it, the rate at which an edge appears where the reference has none and the retention of the weakest structures, both in Section V-J. Neither cost is zero, and the criterion that chose the operating point scores six measures and fidelity and sees neither. We measured that blindness on validation, so the test split is spent once. The runner up setting for NAFNet keeps more weak structure, 43.9 percent against 37.7, and invents fewer edges, so a user who values faint detail should score these two during selection. We did not reselect, the criterion having been fixed before the test split was scored. The measures are established proxies and no clinician read the images, so Section V-J is quantitative evidence and not clinical validation. The external sets are small, sixteen and eighteen images, so the transfer results show behavior across scanners rather than a precise estimate for any one device.

## VII. Conclusion

We presented a post-correction layer for OCT denoising, attached to a frozen denoiser rather than replacing it. It states six clinical properties in closed form and turns them into an inspectable per-pixel allocation for the multiplicative gain through five named rules, alongside an edge correction and a background operation that do not pass through that allocation, and bounds how far any may move the image. Across four backbone families the same rules, budgets and weights raise the clinical measures at a fidelity cost below one decibel, and on three of the four no measure falls. The measures are proxies, and a reader study is the next step.

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